Solutions: Stochastic Finance & Risk Modelling Final Exam

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1 Exercise 1. Chooser Option

1. **Fundamental Theorem of Asset Pricing (FTAP)**

   In a finite discrete-time market, the market is arbitrage-free if and only if there exists an Equivalent Martingale Measure (EMM), denoted by $\mathbb{P}^*$, such that the discounted asset price process $(\tilde{S}_n)_{n\in\mathbb{N}}$ where $\tilde{S}_n = e^{-rn}S_n$ is a martingale under $\mathbb{P}^*$.

   Furthermore, the market is complete if and only if this Equivalent Martingale Measure $\mathbb{P}^*$ is unique. Under this unique EMM, the arbitrage-free price V_n at time n of any attainable $\mathcal{F}_N$-measurable payoff X at maturity N is given by the conditional expectation of its discounted payoff:

   $$V_n = \mathbb{E}^*\left[e^{-r(N-n)}X \mid \mathcal{F}_n\right]$$

2. **Arbitrage-free price at T_c**

   At the decision date T_c, the buyer can choose whether the option becomes a Call or a Put. A rational investor will choose the option with the highest value at that time.

   The payoff at time N is given by $(S_N - K)_+\mathbb{1}_{C_{T_c}\geq P_{T_c}} + (K - S_N)_+\mathbb{1}_{C_{T_c}<P_{T_c}}$.

   Using the risk-neutral pricing formula from time T_c, the price of the chooser option at T_c, let's denote it V_{T_c}, is:

   $$V_{T_c} = \mathbb{E}^*\left[e^{-r(N-T_c)}\left((S_N - K)_+\mathbb{1}_{C_{T_c}\geq P_{T_c}} + (K - S_N)_+\mathbb{1}_{C_{T_c}<P_{T_c}}\right) \mid \mathcal{F}_{T_c}\right]$$

   Since the indicator functions $\mathbb{1}_{C_{T_c}\geq P_{T_c}}$ and $\mathbb{1}_{C_{T_c}<P_{T_c}}$ are $\mathcal{F}_{T_c}$-measurable, we can take them out of the conditional expectation:

   $$V_{T_c} = \mathbb{1}_{C_{T_c}\geq P_{T_c}}\mathbb{E}^*\left[e^{-r(N-T_c)}(S_N - K)_+ \mid \mathcal{F}_{T_c}\right] + \mathbb{1}_{C_{T_c}<P_{T_c}}\mathbb{E}^*\left[e^{-r(N-T_c)}(K - S_N)_+ \mid \mathcal{F}_{T_c}\right]$$

   Recognizing the pricing formulas for the Call and Put options at time T_c:

   $$C_{T_c} = \mathbb{E}^*\left[e^{-r(N-T_c)}(S_N - K)_+ \mid \mathcal{F}_{T_c}\right]$$

   $$P_{T_c} = \mathbb{E}^*\left[e^{-r(N-T_c)}(K - S_N)_+ \mid \mathcal{F}_{T_c}\right]$$

   We get:

   $$V_{T_c} = \mathbb{1}_{C_{T_c}\geq P_{T_c}}C_{T_c} + \mathbb{1}_{C_{T_c}<P_{T_c}}P_{T_c}$$

   This expression is exactly the definition of the maximum between C_{T_c} and P_{T_c}. Thus, $V_{T_c} = \max(C_{T_c}, P_{T_c})$.

3. **Price at T_c using Put-Call parity**

   The Put-Call parity relation at time t for European options on a non-dividend-paying stock is given by:

   $$C_t - P_t = S_t - e^{-r(N-t)}K$$

   Rearranging this formula to express the Put price in terms of the Call price at time T_c:

   $$P_{T_c} = C_{T_c} - S_{T_c} + e^{-r(N-T_c)}K$$

   Now, substitute this expression for P_{T_c} into the result from question 2:

   $$V_{T_c} = \max(C_{T_c}, P_{T_c}) = \max\left(C_{T_c}, C_{T_c} - S_{T_c} + e^{-r(N-T_c)}K\right)$$

   We can factor out C_{T_c} from the maximum function:

   $$V_{T_c} = C_{T_c} + \max\left(0, e^{-r(N-T_c)}K - S_{T_c}\right)$$

   This is equivalent to:

   $$V_{T_c} = C_{T_c} + \left(e^{-r(N-T_c)}K - S_{T_c}\right)_+$$

4. **Replication portfolio**

   From the previous question, the value of the "chooser" option at time T_c is $C_{T_c} + (e^{-r(N-T_c)}K - S_{T_c})_+$. This value can be viewed as the payoff of a specific portfolio at time T_c.

   Let's break down the two terms:

   - C_{T_c} is the value at time T_c of a standard European Call option with strike K and maturity N.
   - $(e^{-r(N-T_c)}K - S_{T_c})_+$ is exactly the payoff at time T_c of a standard European Put option with strike $K' = e^{-r(N-T_c)}K$ and maturity T_c.

   Therefore, to replicate the payoff of the chooser option at T_c, an investor can buy a portfolio at time 0 consisting of:

   - One European Call option with strike K and maturity N.
   - One European Put option with strike $e^{-r(N-T_c)}K$ and maturity T_c.

5. **Arbitrage-free price at 0**

   By the principle of no-arbitrage, if two portfolios have the same value at a future date T_c in all states of the world, they must have the same value today (time 0).

   Since the chooser option can be perfectly replicated by the portfolio described in question 4, its price at time 0 is simply the sum of the prices of the components of that replicating portfolio evaluated at time 0.

   Using the given notation:

   - The price at 0 of the Call option with strike K and maturity N is $C_0(K, N)$.
   - The price at 0 of the Put option with strike $e^{-r(N-T_c)}K$ and maturity T_c is $P_0(e^{-r(N-T_c)}K, T_c)$.

   Thus, the arbitrage-free price of the chooser option at time 0 is:

$$V_0 = C_0(K, N) + P_0\left(e^{-r(N-T_c)}K, T_c\right)$$

2 Exercise 2. American/European Option

1. **Recursive formulation of $\mathcal{C}_n$**

   Let $\mathbb{P}^*$ be the unique Equivalent Martingale Measure (risk-neutral measure) in this complete arbitrage-free market. The arbitrage-free value of an American option $\mathcal{C}_n$ at time n is given by the Snell envelope of the discounted payoff process. The recursive formulation (backward induction) is:

   At maturity N:

$$\mathcal{C}_N = Z_N$$

   For $n \in \{0, \ldots, N-1\}$:

$$\mathcal{C}_n = \max\left(Z_n, \frac{1}{1+r}\mathbb{E}^*\left[\mathcal{C}_{n+1} \mid \mathcal{F}_n\right]\right)$$

   Here, Z_n represents the intrinsic value (immediate exercise value), and $\frac{1}{1+r}\mathbb{E}^*[\mathcal{C}_{n+1} \mid \mathcal{F}_n]$ represents the continuation value.

2. **Showing that $\mathcal{C}_n \geq c_n$**

   The European option value c_n at time n is the expected discounted payoff at maturity N:

$$c_n = \mathbb{E}^*\left[\frac{1}{(1+r)^{N-n}}Z_N \mid \mathcal{F}_n\right]$$

The American option gives the holder the right to exercise at any stopping time τ taking values in $\{n, \ldots, N\}$. Its value is the supremum over all such stopping times:

$$\mathcal{C}_n = \sup_{\tau \in \mathcal{T}_{n,N}} \mathbb{E}^* \left[\frac{1}{(1+r)^{\tau-n}} Z_\tau \mid \mathcal{F}_n \right]$$

Since choosing to hold the option until maturity ($\tau = N$) is one possible valid stopping time in the set $\mathcal{T}_{n,N}$, the supremum over all possible stopping times must be greater than or equal to the expected value of this specific strategy:

$$\mathcal{C}_n \geq \mathbb{E}^* \left[\frac{1}{(1+r)^{N-n}} Z_N \mid \mathcal{F}_n \right] = c_n$$

Therefore, $\mathcal{C}_n \geq c_n$ for all $n = 0 \ldots N$.

3. **Showing that if $c_n \geq Z_n$ then $\mathcal{C}_n = c_n$**

   We will prove this by backward induction on n.

   Base case at $n = N$: By definition, $\mathcal{C}_N = Z_N$ and $c_N = Z_N$. Thus, $\mathcal{C}_N = c_N$.

   Inductive step: Assume that for a given $n + 1 \leq N$, we have $\mathcal{C}_{n+1} = c_{n+1}$. We need to show that $\mathcal{C}_n = c_n$.

   Using the recursive formulation for the American option:

$$\mathcal{C}_n = \max \left(Z_n, \frac{1}{1+r} \mathbb{E}^* \left[\mathcal{C}_{n+1} \mid \mathcal{F}_n \right] \right)$$

   By our induction hypothesis, we replace $\mathcal{C}_{n+1}$ with c_{n+1}:

$$\mathcal{C}_n = \max \left(Z_n, \frac{1}{1+r} \mathbb{E}^* \left[c_{n+1} \mid \mathcal{F}_n \right] \right)$$

   We know that the discounted European option price process is a martingale under $\mathbb{P}^*$, which means:

$$c_n = \frac{1}{1+r} \mathbb{E}^* \left[c_{n+1} \mid \mathcal{F}_n \right]$$

   Substituting this back into the maximum equation:

$$\mathcal{C}_n = \max(Z_n, c_n)$$

   The problem statement provides the condition that $c_n \geq Z_n$ for all n. Therefore, the maximum between Z_n and c_n is simply c_n:

$$\mathcal{C}_n = c_n$$

   By induction, $\mathcal{C}_n = c_n$ holds for all $n = 0 \ldots N$.

4. **Application to a European Call option**

   For a call option on a single risky asset S with strike K and maturity N, the payoff is $Z_n = (S_n - K)_+$. We want to check if the condition $c_n \geq Z_n$ holds. The European call price is:

$$c_n = \mathbb{E}^* \left[\frac{1}{(1+r)^{N-n}} (S_N - K)_+ \mid \mathcal{F}_n \right]$$

   By Jensen's inequality, since the function $x \mapsto \max(x, 0)$ is convex:

$$c_n \geq \left(\mathbb{E}^* \left[\frac{S_N - K}{(1+r)^{N-n}} \mid \mathcal{F}_n \right] \right)_+$$

$$c_n \geq \left(\mathbb{E}^* \left[\frac{S_N}{(1+r)^{N-n}} \mid \mathcal{F}_n \right] - \frac{K}{(1+r)^{N-n}} \right)_+$$

Since the discounted stock price is a martingale under $\mathbb{P}^*$, we have $\mathbb{E}^*[\frac{S_N}{(1+r)^{N-n}} \mid \mathcal{F}_n] = S_n$.

$$c_n \geq \left(S_n - \frac{K}{(1+r)^{N-n}} \right)_+$$

Given that the interest rate $r \geq 0$, it follows that $(1+r)^{N-n} \geq 1$, which implies $\frac{K}{(1+r)^{N-n}} \leq K$. Subtracting a smaller number yields a larger result, so:

$$S_n - \frac{K}{(1+r)^{N-n}} \geq S_n - K$$

Applying the positive part operator preserves this inequality:

$$\left(S_n - \frac{K}{(1+r)^{N-n}} \right)_+ \geq (S_n - K)_+$$

Therefore, we conclude:

$$c_n \geq (S_n - K)_+ = Z_n$$

Since $c_n \geq Z_n$ for all n, we can apply the result from question 3 to conclude that $\mathcal{C}_n = c_n$. An American call option on a non-dividend-paying asset has the exact same price as a European call option, meaning it is never optimal to exercise it early.

3 Exercise 3. Optimal Portfolio

1. Compute the optimizer w_ϕ^* as a function of a, b, c and Σ, μ, and ϕ.

   We want to solve the problem:

$$\max_{w \in \mathbb{R}^n} \left(\mu^T w - \frac{\phi}{2} w^T \Sigma w \right) \quad \text{s.t.} \quad \mathbf{1}^T w = 1$$

   We set up the Lagrangian for this optimization problem. Let λ be the Lagrange multiplier associated with the budget constraint:

$$\mathcal{L}(w, \lambda) = \mu^T w - \frac{\phi}{2} w^T \Sigma w + \lambda(1 - \mathbf{1}^T w)$$

   The first-order conditions for an extremum are obtained by setting the gradients with respect to w and λ to zero:

$$\nabla_w \mathcal{L} = \mu - \phi \Sigma w - \lambda \mathbf{1} = 0$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = 1 - \mathbf{1}^T w = 0$$

   From the first equation, assuming Σ is invertible, we can solve for w:

$$\phi \Sigma w = \mu - \lambda \mathbf{1} \implies w = \frac{1}{\phi} \Sigma^{-1} \mu - \frac{\lambda}{\phi} \Sigma^{-1} \mathbf{1}$$

   Substitute this expression for w into the second first-order condition (the constraint $\mathbf{1}^T w = 1$):

$$\mathbf{1}^T \left(\frac{1}{\phi} \Sigma^{-1} \mu - \frac{\lambda}{\phi} \Sigma^{-1} \mathbf{1} \right) = 1$$

$$\frac{1}{\phi}\mathbf{1}^T\Sigma^{-1}\mu - \frac{\lambda}{\phi}\mathbf{1}^T\Sigma^{-1}\mathbf{1} = 1$$

Using the definitions $a = \mu^T\Sigma^{-1}\mathbf{1} = \mathbf{1}^T\Sigma^{-1}\mu$ (since Σ^{-1} is symmetric) and $c = \mathbf{1}^T\Sigma^{-1}\mathbf{1}$, we get:

$$\frac{a}{\phi} - \frac{\lambda c}{\phi} = 1 \implies a - \lambda c = \phi \implies \lambda = \frac{a-\phi}{c}$$

Now substitute λ back into the expression for w to find the optimal portfolio weights w_ϕ^*:

$$\begin{aligned} w_\phi^* &= \frac{1}{\phi}\Sigma^{-1}\mu - \frac{1}{\phi}\left(\frac{a-\phi}{c}\right)\Sigma^{-1}\mathbf{1} \\ w_\phi^* &= \frac{1}{\phi}\Sigma^{-1}\mu - \frac{a-\phi}{\phi c}\Sigma^{-1}\mathbf{1} \end{aligned}$$

2. Deduce the limit $\lim_{\phi\to\infty} w_\phi^*$.

   Taking the limit of the expression found in question 1 as ϕ goes to infinity:

$$\begin{aligned} \lim_{\phi\to\infty} w_\phi^* &= \lim_{\phi\to\infty}\left(\frac{1}{\phi}\Sigma^{-1}\mu - \frac{a-\phi}{\phi c}\Sigma^{-1}\mathbf{1}\right) \\ &= \lim_{\phi\to\infty}\frac{1}{\phi}\Sigma^{-1}\mu - \lim_{\phi\to\infty}\left(\frac{a}{\phi c} - \frac{1}{c}\right)\Sigma^{-1}\mathbf{1} \\ &= 0 - \left(0 - \frac{1}{c}\right)\Sigma^{-1}\mathbf{1} \\ &= \frac{1}{c}\Sigma^{-1}\mathbf{1} \end{aligned}$$

   This represents the global minimum variance portfolio, which is the optimal portfolio for an infinitely risk-averse investor.

3. Compute the return and the variance of w_ϕ^*.

   The expected return R_ϕ is given by $\mu^T w_\phi^*$:

$$\begin{aligned} R_\phi &= \mu^T\left(\frac{1}{\phi}\Sigma^{-1}\mu - \frac{a-\phi}{\phi c}\Sigma^{-1}\mathbf{1}\right) \\ &= \frac{1}{\phi}\mu^T\Sigma^{-1}\mu - \frac{a-\phi}{\phi c}\mu^T\Sigma^{-1}\mathbf{1} \\ &= \frac{b}{\phi} - \frac{a(a-\phi)}{\phi c} \\ &= \frac{bc - a^2 + a\phi}{\phi c} = \frac{bc-a^2}{\phi c} + \frac{a}{c} \end{aligned}$$

The variance V_ϕ is given by $(w_\phi^*)^T \Sigma w_\phi^*$. We use the expanded form of w_ϕ^*:

$$\begin{aligned}
V_\phi &= \left(\frac{1}{\phi}\mu^T\Sigma^{-1} - \frac{a-\phi}{\phi c}\mathbf{1}^T\Sigma^{-1}\right)\Sigma\left(\frac{1}{\phi}\Sigma^{-1}\mu - \frac{a-\phi}{\phi c}\Sigma^{-1}\mathbf{1}\right) \\
&= \left(\frac{1}{\phi}\mu^T - \frac{a-\phi}{\phi c}\mathbf{1}^T\right)\left(\frac{1}{\phi}\Sigma^{-1}\mu - \frac{a-\phi}{\phi c}\Sigma^{-1}\mathbf{1}\right) \\
&= \frac{1}{\phi^2}\mu^T\Sigma^{-1}\mu - \frac{a-\phi}{\phi^2 c}\mu^T\Sigma^{-1}\mathbf{1} - \frac{a-\phi}{\phi^2 c}\mathbf{1}^T\Sigma^{-1}\mu + \frac{(a-\phi)^2}{\phi^2 c^2}\mathbf{1}^T\Sigma^{-1}\mathbf{1} \\
&= \frac{b}{\phi^2} - \frac{a(a-\phi)}{\phi^2 c} - \frac{a(a-\phi)}{\phi^2 c} + \frac{(a-\phi)^2}{\phi^2 c^2}c \\
&= \frac{bc - 2a^2 + 2a\phi + a^2 - 2a\phi + \phi^2}{\phi^2 c} \\
&= \frac{bc - a^2 + \phi^2}{\phi^2 c} = \frac{bc - a^2}{\phi^2 c} + \frac{1}{c}
\end{aligned}$$

4. Prove that $bc - a^2 > 0$ and compare the return of w_ϕ^* to a/c.

   Since Σ is positive definite, Σ^{-1} is also positive definite. We can define an inner product on $\mathbb{R}^n$ as $\langle x, y\rangle = x^T\Sigma^{-1}y$. Applying the Cauchy-Schwarz inequality to the vectors μ and $\mathbf{1}$:

$$\begin{aligned}
\langle \mu, \mathbf{1}\rangle^2 &\leq \langle \mu, \mu\rangle\langle \mathbf{1}, \mathbf{1}\rangle \\
(\mu^T\Sigma^{-1}\mathbf{1})^2 &\leq (\mu^T\Sigma^{-1}\mu)(\mathbf{1}^T\Sigma^{-1}\mathbf{1}) \\
a^2 &\leq b \cdot c
\end{aligned}$$

   The problem states that μ is not collinear to $\mathbf{1}$. Thus, the vectors are linearly independent, making the Cauchy-Schwarz inequality strict:

$$a^2 < bc \implies bc - a^2 > 0$$

   Now comparing the return R_ϕ to a/c:

$$R_\phi = \frac{bc - a^2}{\phi c} + \frac{a}{c}$$

   Since $bc - a^2 > 0$, and we know $\phi > 0$ and $c > 0$ (because Σ^{-1} is positive definite and $c = \mathbf{1}^T\Sigma^{-1}\mathbf{1}$), the term $\frac{bc-a^2}{\phi c}$ is strictly positive. Therefore, $R_\phi > \frac{a}{c}$.

5. Express the variance of w_ϕ^* as a function of its return. What do you deduce?

   From the expression for return derived in question 3, we can isolate $\frac{1}{\phi}$:

$$R_\phi - \frac{a}{c} = \frac{bc - a^2}{\phi c} \implies \frac{1}{\phi} = \frac{c\left(R_\phi - \frac{a}{c}\right)}{bc - a^2}$$

   Now substitute this into the variance equation:

$$\begin{aligned}
V_\phi &= \frac{bc - a^2}{c}\left(\frac{1}{\phi}\right)^2 + \frac{1}{c} \\
&= \frac{bc - a^2}{c}\left[\frac{c\left(R_\phi - \frac{a}{c}\right)}{bc - a^2}\right]^2 + \frac{1}{c} \\
&= \frac{c(R_\phi - a/c)^2}{bc - a^2} + \frac{1}{c}
\end{aligned}$$

   We deduce that the relationship between the variance V_ϕ and the expected return R_ϕ describes a parabola in the variance-return plane. This equation represents the efficient frontier (the Markowitz bullet) in modern portfolio theory. The minimum possible variance is $1/c$, which occurs when the expected return is a/c (this aligns with the global minimum variance portfolio found when $\phi \to \infty$).

4 Exercise 4. Markov Process and Average Options

4.1 I) Introduction

1. **Show that $\mathcal{F}_n = \sigma(S_1, \ldots, S_n)$ and its meaning.**

   By definition, $\mathcal{F}_n = \sigma(U_1, \ldots, U_n)$. We know that $S_k = S_0 \prod_{i=1}^{k} U_i$ for any $k \geq 1$. Since S_0 is a strictly positive constant, each S_k is a continuous function of $U_1, \ldots, U_k$, meaning S_k is measurable with respect to $\sigma(U_1, \ldots, U_k)$. Therefore, $\sigma(S_1, \ldots, S_n) \subseteq \sigma(U_1, \ldots, U_n) = \mathcal{F}_n$.

   Conversely, we can recover U_k from the prices since $U_k = \frac{S_k}{S_{k-1}}$ for $k \geq 1$. Thus, each U_k is a measurable function of S_{k-1} and S_k, which means U_k is measurable with respect to $\sigma(S_1, \ldots, S_n)$. Therefore, $\mathcal{F}_n = \sigma(U_1, \ldots, U_n) \subseteq \sigma(S_1, \ldots, S_n)$.

   We conclude that $\mathcal{F}_n = \sigma(S_1, \ldots, S_n)$. Economically, $\mathcal{F}_n$ represents the information available to investors at time n, which is exactly the entire history of the risky asset's prices up to that time.

2. **Condition for an arbitrage-free and complete market.**

   In the standard binomial model, the necessary and sufficient condition for the market to be arbitrage-free and complete is:

   $$d < 1 + r < u$$

3. **Defining $\mathbb{P}^*$ and the behavior of (U_n).**

   For the discounted price process $\hat{S}_n = \frac{S_n}{(1+r)^n}$ to be a martingale under $\mathbb{P}^*$, we need $\mathbb{E}^*[\hat{S}_{n+1} \mid \mathcal{F}_n] = \hat{S}_n$.

   This implies $\mathbb{E}^*\left[\frac{S_n U_{n+1}}{(1+r)^{n+1}} \mid \mathcal{F}_n\right] = \frac{S_n}{(1+r)^n}$.

   Since S_n is $\mathcal{F}_n$-measurable, it can be taken out of the expectation:

   $$\frac{S_n}{(1+r)^{n+1}} \mathbb{E}^*[U_{n+1} \mid \mathcal{F}_n] = \frac{S_n}{(1+r)^n} \implies \mathbb{E}^*[U_{n+1} \mid \mathcal{F}_n] = 1 + r$$

   Let $p^* = \mathbb{P}^*(U_{n+1} = u \mid \mathcal{F}_n)$. Since U_{n+1} only takes values u and d:

   $$p^* u + (1 - p^*) d = 1 + r$$

   Solving for p^*:

   $$p^*(u - d) + d = 1 + r \implies p^* = \frac{1 + r - d}{u - d}$$

   Under $\mathbb{P}^*$, the random variables $(U_n)_{n \in \{1, \ldots, N\}}$ are independent and identically distributed (i.i.d.), each taking the value u with probability p^* and d with probability $1 - p^*$.

4. **Proving the conditional expectation property.**

   Let g be a Borel measurable and integrable function. We want to evaluate $\mathbb{E}^*[g(S_1, \ldots, S_n, U_{n+1}) \mid \mathcal{F}_n]$.

   The random variables $S_1, \ldots, S_n$ are $\mathcal{F}_n$-measurable. The random variable U_{n+1} is independent of $\mathcal{F}_n$ under the probability measure $\mathbb{P}^*$ (since U_i are i.i.d.).

   By the Independence Lemma for conditional expectations (also known as the Freezing Lemma), if we have a function $g(X, Y)$ where X is $\mathcal{F}$-measurable and Y is independent of $\mathcal{F}$, then $\mathbb{E}[g(X, Y) \mid \mathcal{F}] = G(X)$, where $G(x) = \mathbb{E}[g(x, Y)]$.

   Applying this directly with $X = (S_1, \ldots, S_n)$ and $Y = U_{n+1}$:

   $$\mathbb{E}^*[g(S_1, \ldots, S_n, U_{n+1}) \mid \mathcal{F}_n] = G(S_1, \ldots, S_n)$$

   where $G(s_1, \ldots, s_n) = \mathbb{E}^*[g(s_1, \ldots, s_n, U_{n+1})]$.

4.2 II) Option Pricing

1. **Evaluating the conditional expectation for $h(S_{n+1}, \Sigma_{n+1})$.**

   We have $S_{n+1} = S_n U_{n+1}$ and $\Sigma_{n+1} = \Sigma_n + S_{n+1} = \Sigma_n + S_n U_{n+1}$.

   Therefore, $h(S_{n+1}, \Sigma_{n+1}) = h(S_n U_{n+1}, \Sigma_n + S_n U_{n+1})$. Let's define a function $g(S_n, \Sigma_n, U_{n+1}) = h(S_n U_{n+1}, \Sigma_n + S_n U_{n+1})$.

   Using the result from I.4, since S_n and Σ_n are $\mathcal{F}_n$-measurable and U_{n+1} is independent of $\mathcal{F}_n$:

   $$\mathbb{E}^*[h(S_n U_{n+1}, \Sigma_n + S_n U_{n+1}) \mid \mathcal{F}_n] = G(S_n, \Sigma_n)$$

   where $G(x, y) = \mathbb{E}^*[h(xU_{n+1}, y + xU_{n+1})]$.

   To compute this expectation over U_{n+1}, we use its distribution under $\mathbb{P}^*$ (let $\pi = p^* = \frac{1+r-d}{u-d}$):

   $$G(x, y) = \pi h(xu, y + xu) + (1 - \pi) h(xd, y + xd)$$

   Substituting S_n for x and Σ_n for y yields:

   $$\mathbb{E}^*[h(S_{n+1}, \Sigma_{n+1}) \mid \mathcal{F}_n] = \pi h(uS_n, \Sigma_n + uS_n) + (1 - \pi) h(dS_n, \Sigma_n + dS_n)$$

2. **Deducing the Markovian property of C_n.**

   A process $C = (C_n)_{n \in \{0, \ldots, N\}}$ is a Markov process under $\mathbb{P}^*$ with respect to the filtration $\mathbb{F}$ if, for any bounded measurable function h, $\mathbb{E}^*[h(C_{n+1}) \mid \mathcal{F}_n] = \mathbb{E}^*[h(C_{n+1}) \mid C_n]$.

   From the result in II.1, for any function h, the conditional expectation $\mathbb{E}^*[h(C_{n+1}) \mid \mathcal{F}_n]$ depends on the past history only through the values of S_n and Σ_n.

   Since $C_n = (S_n, \Sigma_n)$, we have $\mathbb{E}^*[h(C_{n+1}) \mid \mathcal{F}_n] = f(C_n)$ for some function f. This confirms that C_n is an $(\mathbb{F}, \mathbb{P}^*)$-Markovian process.

3. **Arbitrage-free value of the average option and its recursive form.**

   By the Fundamental Theorem of Asset Pricing, the arbitrage-free value E_n at time n of a European option with payoff $e(S_N, \Sigma_N)$ at maturity N is the discounted expected payoff under the risk-neutral measure $\mathbb{P}^*$:

   $$E_n = \mathbb{E}^* \left[\frac{1}{(1+r)^{N-n}} e(S_N, \Sigma_N) \mid \mathcal{F}_n \right]$$

   We can also express this recursively. At maturity N, the value is simply the payoff:

   $$E_N = e(S_N, \Sigma_N)$$

   So we can define $e_N(x, y) = e(x, y)$.

   For any time $n < N$, the value is the discounted expected value of the option at time $n + 1$:

   $$E_n = \frac{1}{1+r} \mathbb{E}^*[E_{n+1} \mid \mathcal{F}_n]$$

   Assume by backward induction that at time $n + 1$, $E_{n+1} = e_{n+1}(S_{n+1}, \Sigma_{n+1})$ for some function e_{n+1}. Then:

   $$E_n = \frac{1}{1+r} \mathbb{E}^*[e_{n+1}(S_{n+1}, \Sigma_{n+1}) \mid \mathcal{F}_n]$$

   Applying the result from II.1 to the function e_{n+1}:

   $$E_n = \frac{1}{1+r} \left[\pi e_{n+1}(uS_n, \Sigma_n + uS_n) + (1 - \pi) e_{n+1}(dS_n, \Sigma_n + dS_n) \right]$$

   This shows that E_n is a function of only S_n and Σ_n. We can define this function $e_n(x, y)$ as:

   $$e_n(x, y) = \frac{1}{1+r} \left[\pi e_{n+1}(ux, xu + y) + (1 - \pi) e_{n+1}(dx, xd + y) \right]$$

   Thus, $E_n = e_n(S_n, \Sigma_n)$, proving the required recursive relationship.

5 Exercise 5. Risk Management

1. *Compute the loss $L(x,p)$.*

   The initial value of the portfolio at time 0 is given by:

$$V_0 = xS_0 + yS_0^0 - p$$

   Since $S_0 = s$, $S_0^0 = 1$, and (x,y) is a hedging strategy such that $p = xs + y$, we can substitute y:

$$V_0 = xs + (p - xs) - p = 0$$

   The value of the portfolio at time 1 is:

$$V_1 = xS_1 + yS_1^0 - F(S_1)$$

   Since $S_1^0 = 1$, $S_1 = s + \Delta S$, and $y = p - xs$:

$$V_1 = x(s + \Delta S) + p - xs - F(S_1) = p + x\Delta S - F(S_1)$$

   The loss $L(x,p)$ is defined as $V_0 - V_1$:

$$L(x,p) = 0 - (p + x\Delta S - F(S_1)) = F(S_1) - p - x\Delta S$$

2. *Optimality conditions and solving for p^*.*

   We want to minimize the risk measure $\rho(L(x,p)) = \mathbb{E}[L(x,p)^2] = \mathbb{E}[(F(S_1) - p - x\Delta S)^2]$. The first-order optimality conditions require the partial derivatives with respect to p and x to be zero:

   With respect to p:

$$\frac{\partial}{\partial p}\mathbb{E}[(F(S_1) - p - x\Delta S)^2] = \mathbb{E}[-2(F(S_1) - p - x\Delta S)] = 0$$

$$\mathbb{E}[F(S_1)] - p - x\mathbb{E}[\Delta S] = 0 \implies p = \mathbb{E}[F(S_1)] - x\mathbb{E}[\Delta S]$$

   With respect to x:

$$\frac{\partial}{\partial x}\mathbb{E}[(F(S_1) - p - x\Delta S)^2] = \mathbb{E}[-2\Delta S(F(S_1) - p - x\Delta S)] = 0$$

$$\mathbb{E}[F(S_1)\Delta S] - p\mathbb{E}[\Delta S] - x\mathbb{E}[\Delta S^2] = 0$$

   Substitute the expression for p into this second equation:

$$\mathbb{E}[F(S_1)\Delta S] - (\mathbb{E}[F(S_1)] - x\mathbb{E}[\Delta S])\mathbb{E}[\Delta S] - x\mathbb{E}[\Delta S^2] = 0$$

$$\mathbb{E}[F(S_1)\Delta S] - \mathbb{E}[F(S_1)]\mathbb{E}[\Delta S] + x(\mathbb{E}[\Delta S])^2 - x\mathbb{E}[\Delta S^2] = 0$$

   Factor out x and recognize that $\mathbb{E}[\Delta S^2] - (\mathbb{E}[\Delta S])^2 = \mathrm{var}(\Delta S)$:

$$x \cdot \mathrm{var}(\Delta S) = \mathbb{E}[F(S_1)\Delta S] - \mathbb{E}[F(S_1)]\mathbb{E}[\Delta S]$$

$$x^* = \frac{\mathbb{E}[F(S_1)\Delta S] - \mathbb{E}[F(S_1)]\mathbb{E}[\Delta S]}{\mathrm{var}(\Delta S)}$$

   Now, plug x^* back into the equation for p:

$$p^* = \mathbb{E}[F(S_1)] - \left(\frac{\mathbb{E}[F(S_1)\Delta S] - \mathbb{E}[F(S_1)]\mathbb{E}[\Delta S]}{\mathrm{var}(\Delta S)}\right)\mathbb{E}[\Delta S]$$

$$p^* = \frac{\mathbb{E}[F(S_1)]\mathrm{var}(\Delta S) - \mathbb{E}[F(S_1)\Delta S]\mathbb{E}[\Delta S] + \mathbb{E}[F(S_1)](\mathbb{E}[\Delta S])^2}{\mathrm{var}(\Delta S)}$$

   Since $\mathrm{var}(\Delta S) + (\mathbb{E}[\Delta S])^2 = \mathbb{E}[\Delta S^2]$, we can combine the terms with $\mathbb{E}[F(S_1)]$:

$$p^* = \frac{\mathbb{E}[\Delta S^2]\mathbb{E}[F(S_1)] - \mathbb{E}[\Delta S]\mathbb{E}[F(S_1)\Delta S]}{\mathrm{var}(\Delta S)}$$

3. *Assuming S is a martingale under $\mathbb{P}$.*

   If S is a martingale, then $\mathbb{E}[S_1 \mid \mathcal{F}_0] = S_0$. This implies $\mathbb{E}[\Delta S] = 0$. Substituting $\mathbb{E}[\Delta S] = 0$ into the formula for p^*:

$$p^* = \frac{\mathbb{E}[\Delta S^2]\mathbb{E}[F(S_1)] - 0}{\mathrm{var}(\Delta S)}$$

   Since $\mathbb{E}[\Delta S] = 0$, we have $\mathrm{var}(\Delta S) = \mathbb{E}[\Delta S^2]$. Therefore:

$$p^* = \frac{\mathbb{E}[\Delta S^2]\mathbb{E}[F(S_1)]}{\mathbb{E}[\Delta S^2]} = \mathbb{E}[F(S_1)]$$

   Because the risk-free asset is constant ($S^0 \equiv 1$), the interest rate $r = 0$. Furthermore, since S is a martingale under $\mathbb{P}$, $\mathbb{P}$ is the risk-neutral measure. The arbitrage-free price is precisely the expected payoff under the risk-neutral measure discounted at rate $r = 0$, which is $\mathbb{E}[F(S_1)]$. Thus, p^* is exactly equal to the arbitrage-free price.

4. *Compute p^* in the binomial model context.*

   Let $F_u = F((1+u)s)$ and $F_d = F((1+d)s)$. We know ΔS takes values us with probability q and ds with probability $1-q$.

$$\begin{aligned}
\mathbb{E}[\Delta S] &= q(us) + (1-q)(ds) = s(qu + (1-q)d) \\
\mathbb{E}[\Delta S^2] &= q(u^2s^2) + (1-q)(d^2s^2) = s^2(qu^2 + (1-q)d^2) \\
\mathbb{E}[F(S_1)] &= qF_u + (1-q)F_d \\
\mathbb{E}[F(S_1)\Delta S] &= qF_u(us) + (1-q)F_d(ds) = s(quF_u + (1-q)dF_d)
\end{aligned}$$

   Using the hint $\mathrm{var}(\Delta S) = s^2q(1-q)(u-d)^2$, we evaluate the coefficient for F_u in the p^* formula:

$$\frac{qs^2(qu^2 + (1-q)d^2) - s(qu + (1-q)d) \cdot qus}{s^2q(1-q)(u-d)^2}$$

$$= \frac{q(qu^2 + (1-q)d^2 - qu^2 - (1-q)ud)}{q(1-q)(u-d)^2} = \frac{q(1-q)d(d-u)}{q(1-q)(u-d)^2} = \frac{-d}{u-d}$$

   Similarly, evaluating the coefficient for F_d:

$$\frac{(1-q)s^2(qu^2 + (1-q)d^2) - s(qu + (1-q)d) \cdot (1-q)ds}{s^2q(1-q)(u-d)^2}$$

$$= \frac{(1-q)(qu^2 + (1-q)d^2 - qud - (1-q)d^2)}{q(1-q)(u-d)^2} = \frac{(1-q)qu(u-d)}{q(1-q)(u-d)^2} = \frac{u}{u-d}$$

   Summing these parts gives:

$$p^* = -\frac{d}{u-d}F((1+u)s) + \frac{u}{u-d}F((1+d)s)$$

   This matches the arbitrage-free price. In a binomial model with $r = 0$, the risk-neutral probabilities are $\pi_u = \frac{0-d}{u-d}$ and $\pi_d = \frac{u-0}{u-d}$. The arbitrage-free price is $\pi_u F_u + \pi_d F_d$, which is exactly what we derived for p^*.

5. *Perfect Hedging.*

   In the binomial model, the market is complete. We can define a strategy $(\overline{x}, \overline{p})$ that perfectly replicates the derivative. Let $\overline{x} = \frac{F_u - F_d}{s(u-d)}$ and $\overline{p} = p^*$.

   For the up state: $L = F_u - p^* - \overline{x}(us) = 0$ For the down state: $L = F_d - p^* - \overline{x}(ds) = 0$

   Since the loss $L(\overline{x}, \overline{p})$ is identically 0 in all states of the world, its risk measure $\rho(L) = \mathbb{E}[0^2] = 0$.

   Comment: In a complete market, quadratic risk minimization & perfect replication are equivalent. The minimal risk is zero, and the optimal initial capital p^* required for this strategy uniquely aligns with the arbitrage-free price.